

A study on tactile illusions of temperature and touch through virtual hand illusion

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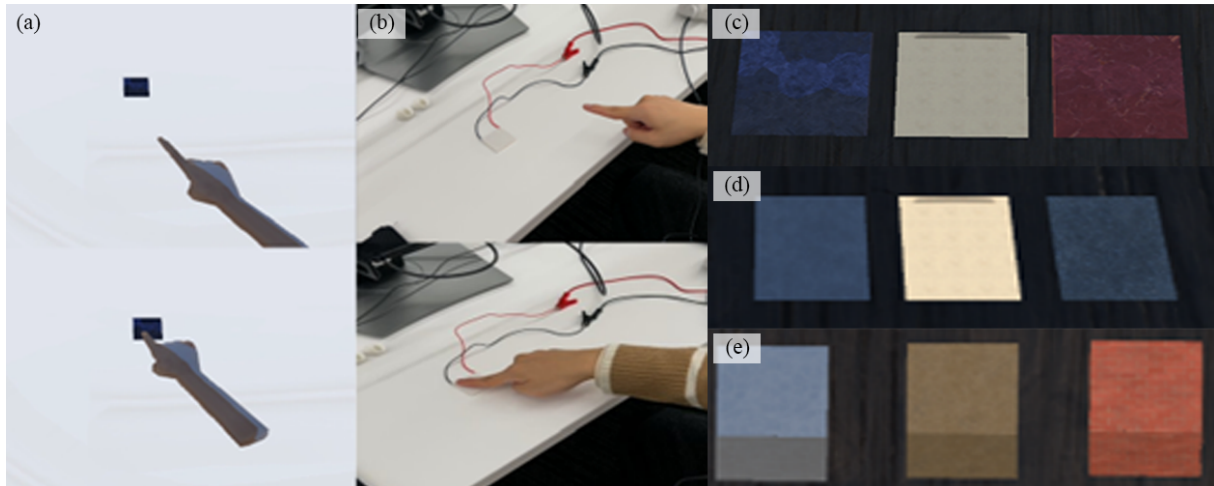


Figure 1: An experiment in virtual hand illusion. (a) A 3D object of a hand displayed in virtual space in synchronization with the real world. (b) Hand motion in the real world (synchronized with (a)). (c)-(e) are 3D objects placed in the virtual space, representing temperature (from left to right: 0°C, 20°C, 40°C), smoothness (from left to right: smooth, medium, rough), and hardness (from left to right: soft, medium, hard), respectively.

ABSTRACT

Recently, a wide variety of VR content has been provided, and the development of devices that approach the sense of touch is attracting attention. We investigated the sensory experience of "temperature," "smoothness," and "hardness" in virtual space. Unlike previous studies that perform haptic feedback using a glove-type device, we conducted a tactile experiment without using a device by inducing a tactile illusion from visual information using a virtual hand illusion. The results suggest that visual information can evoke a tactile illusion of "temperature" and "smoothness."

CCS CONCEPTS

• **Computing methodologies** → **Virtual reality**; • **Human-centered computing** → **Virtual reality**.

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KEYWORDS

VR, crossmodal, virtual hand illusion

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1 INTRODUCTION

In recent years, research on VR (Virtual Reality) has been active. VR is a technology that immerses users in a virtual space by providing a sense of presence close to reality, and advances in communication technology and tactile feedback technology have attracted attention to the development of devices that achieve a higher level of reality.

In conventional research, tactile feedback is provided by wearing a glove-type controller. However, wearing a glove-type controller places significant restrictions on the user. If it is possible to experience tactile sensation without wearing a controller, the restrictions caused by wearing the device can be improved. One way to solve the problem of wearing special devices is to use rubber hand illusions to generate tactile illusions from visual information in a virtual space [Xiong et al. 2021; Yuan and Steed 2010]. This uses a

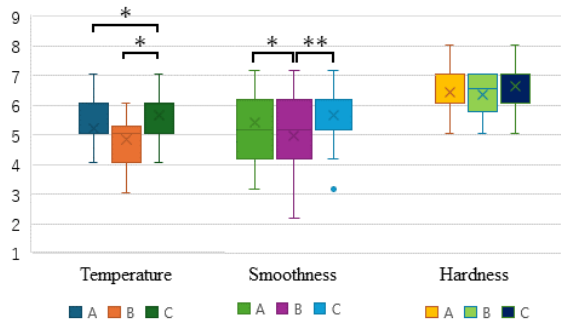


Figure 2: The experiment results.

cross-modal phenomenon between visual and tactile perception, showing that they are closely related to each other. It has also been shown that when 3D objects in virtual space closely resemble objects in the real world in shape, the interaction between virtual space and the real world is natural, and the illusion is more likely to occur when the objects of the illusion are spatially close.

Although methods to experience tactile sensation in virtual space without using a controller have been proposed, research on subdividing the feel has not progressed sufficiently. In this study, we conduct experiments focusing on the sensations of “temperature,” “smoothness,” and “hardness” in a virtual space, using the cross-modal phenomenon of vision and touch.

2 EXPERIMENTS

3D objects representing differences in “temperature,” “hardness,” and “smoothness” are modeled, and subjects touch the 3D objects in a virtual space through an head mounted display. In synchronization with the virtual space, subjects touch objects corresponding to each 3D object in the real world. After the subjects had fully touched the objects corresponding to the 3D objects, they were asked to touch objects with physical quantities different from those in the images to investigate how their senses were changed.

The HMD used in the experiment is a Meta Quest 2, and the VR scene projected on the HMD is created using Unity. Leap Motion Controller 2 is used to track the subject’s hands. The Leap Motion Controller 2 is attached to the end of the arm hand and placed 30 cm above the desk so that the subject could look down on the desk from above. By displaying a CG model of the hand on the coordinates obtained by hand tracking, the position and movement of the hand in the virtual space and in the real world are synchronized.

In the “temperature” experiment, three different temperatures, cold (0°C), warm (40°C), and intermediate (20°C), are reproduced by passing a DC current through a Peltier element using the PS30V5A01 Azwan DC stabilized power supply, and subjects touch three 3D objects representing different temperatures in the order of 0°C, 40°C, and 20°C. Synchronized with the virtual space, the Peltier elements of each temperature are also touched in the real world. The respondents are asked to rate the temperature of 20°C on a 9-point scale, with 1 being cold (0°C) and 9 being warm (40°C). Then, we have subjects wearing HMD repeatedly touch a 3D object representing coldness (0°C) and a 3D object representing warmth (40°C) to learn the object’s temperature. After sufficient learning,

the participants are asked to touch the 3D object representing coldness in the virtual space and the Peltier element set 20°C in the real world, and answer in the same way how much temperature they felt (Figure 1). After having the participants learn the temperature of the 3D object again, they were asked to touch the 3D object representing warmth in the virtual space and the Peltier element at 20°C in the real world, and respond how warm they felt. We compare the initial response (before the illusion) with the response when the images and sensations do not match.

In the “smoothness” experiment, a glossy paper (smooth), a sandpaper (rough), and a Japanese paper (intermediate smoothness) were used, and in the “hardness” experiment, an object with three different physical quantities was prepared using a sponge (soft), plastic (hard), and clay (intermediate hardness), respectively, and the same experiment was conducted as for “temperature”. The virtual 3D objects used in each are shown in Figure 1.

3 RESULTS

20 men and women in their teens and twenties were asked to cooperate in the experiment. 20 responses were obtained for the “temperature” and “smoothness” experiments, and 10 for the “hardness” experiment. The results were compared by Wilcoxon signed rank test. The results for each experiment are shown in Figure 2, with the response before the illusion as A, the response when touching an object with an intermediate physical quantity while looking at a “cold / smooth / soft” 3D object as B, and the response when touching an object with an intermediate physical quantity while looking at a “warm / rough / hard” 3D object as C.

In the “temperature” experiment, there was a significant difference between A and C (p -value: 0.0057), and in the “smoothness” experiment, there was a significant difference between A and B (p -value: 0.0167), indicating the perception of an illusion ($p < 0.05$). No significant differences were found in the “hardness” experiment.

4 CONCLUSIONS AND FUTURE WORK

This study focused on the cross-modal phenomenon between vision and touch, and investigated the perception of touch in virtual space. As a result, it was shown that it is possible to create the illusion of tactile sensation from visual information in terms of “temperature” and “smoothness”. On the other hand, the results showed that it was difficult to perceive the illusion of hardness. As a reason, it was pointed out that it was difficult to imagine the hardness of the 3D model used. Since the 3D model used in this experiment does not have a function to deform by touch, it is possible that the subjects had difficulty imagining the hardness of the model. This is an issue to be addressed in the future.

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